Soft Instance-Level Domain Adaptation With Virtual Classifier for Unsupervised Hyperspectral Image Classification

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***Abstract*— Adversarial learning-based unsupervised hyperspectral image (HSI) classification methods usually adapt probability distributions by minimizing the statistical distance between similar pixels of different HSIs. Since adversarial learning may weaken the discriminability of features, the extracted features will contain a lot of nondiscriminative information, and pixels with similar features may be classified as different classes. Therefore, directly reducing the statistical distance between similar pixels in a latent space may aggravate misclassification. To this end, we propose an unsupervised HSI classification method called soft instance-level domain adaptation with virtual classifier. First, the domain-invariant features of HSI are extracted by a graph convolutional network. Then, a feature similarity metric-based virtual classifier is constructed to output class probabilities of target-domain samples. Furthermore, to enable similar features of HSIs from different domains to be classified into the same class, the divergence between the real and virtual classifiers is reduced by minimizing the real and virtual classifier determinacy disparity. Finally, to reduce the influence of noisy pseudo-labels, a soft instance-level domain adaptation method is proposed. For each target-domain sample, the confidence coefficients are assigned to its corresponding positive and negative samples in the source domain, and a soft prototype contrastive loss is constructed and minimized to adapt two domains in an instance-level way. Experimental results on five real HSI datasets, including Botswana, Kennedy Space Center, Pavia Center, Pavia University, and HyRANK, demonstrate the effectiveness of our proposed method.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image (HSI) classification, instance-level domain adaptation, soft prototype contrastive loss, virtual classifier.**

I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of spectral bands that can represent the reflection properties of ground objects [1]. Compared with multispectral remote sensing images, HSI contains more spectral information [2]. Therefore, HSI has attracted great research interest from the remote sensing community and other fields (e.g., urban development [3], environmental detection [4], [5], agronomy [6], [7], [8], and military [9]). HSI classification aims to assign each pixel with specific class label according to its unique spectral feature and spatial information [10]. However, labeling HSI often requires a lot of time and effort from experts, which allows only few or even no labeled pixels can be utilized for training a classification model [11]. One reasonable solution is to adequately utilize a large amount of labeled and related HSIs captured under different conditions to assist in learning a more accurate classifier. However, traditional classification methods cannot achieve satisfactory results due to the so-called spectral shift, which is caused by the diverse shooting conditions (e.g., atmospheric conditions, soil moisture, and light conditions) [12]. To this end, various domain adaptation methods are applied to the multitemporal and cross-scene HSI classification tasks [13]. Domain adaptation aims to improve the classification accuracy of unlabeled HSIs (target domain) with the help of related but different HSIs (source domain) that contain a large number of labeled pixels.

Deep learning methods have been widely used in domain adaptation due to their powerful feature extraction capability. Zhou and Prasad [14] realized the adaptation of different domains by minimizing a specific ratio, which can simultaneously minimize the within-class distance and maximize the between-class distance in a latent space. Zhang et al. [15] utilized both the latent topological relationships between different classes and the features extracted by convolutional neural networks to construct graphs dynamically. The topological relationships and the feature distributions of different domains were adapted through the graph optimal transmission and maximum mean discrepancy (MMD), respectively. Zhou and Prasad [16] first used a deep convolutional recurrent neural network to extract discriminative features for HSIs, and then, the extracted features were adapted in a layer-by-layer way.

Generative adversarial network [17] has been applied to domain adaptation due to its unique adversarial strategy. Ganin et al. [18] proposed the domain adversarial neural network (DANN) to extract domain-invariant features by an adversarial learning mechanism. However, DANN is not optimal on discriminative tasks and only applicable for small domain shift. Tzeng et al. [19] proposed adversarial

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discriminative domain adaptation by combining adversarial learning with discriminative feature learning. Specifically, discriminative mapping from the target-domain images to the source-domain feature space is learned by fooling a domain discriminator. In view of the outstanding performance, adversarial learning-based domain adaptation methods are also applied to HSI. Elshamli et al. [20] introduced DANN into the remote sensing field and realized the end-to-end HSI unsupervised domain adaptation. Wang et al. [21] performed the domain adversarial learning to reduce the marginal distribution difference and the second-order statistic difference between two domains. Yu et al. [22] achieved the feature alignment with contentwise discrepancy reduction through an adversarial framework. Kong et al. [23] proposed an HSI classification method, which used a graph domain adversarial network and a dual weighted pseudo-label loss to extract more discriminative features and improve the reliability of prediction model.

The above adversarial learning-based domain adaption methods mainly adapt the entire source- and target-domain distributions without considering the complex multimodal structure of data. Recently, some efforts have been made to adapt different domains in a finer way. Liu et al. [24] proposed the classwise distribution adaptation (CDA) for HSI classification. Based on the multiadversarial domain adaptation (MADA) [25], the probabilistic prediction-based MMD method was combined with the class-level adversarial adaptation network to achieve the unsupervised classification of HSI. Tang et al. [26] proposed an unsupervised joint adversarial domain adaptation architecture that considers domain- and class-level feature alignment between source- and target-domain HSIs in a unified adversarial learning process to reduce the distribution difference between different domains. Sharma et al. [27] proposed an instance-level affinity-based domain adversarial learning method. They used the multisample contrastive loss [28] to perform the instance affinity-aware transfer by identifying pairwise similarity relations across the source and target domains. During the adversarial learning procedure, the instance- and class-level data structures may be distorted. Therefore, Zhu et al. [29] proposed to integrate the instance- and class-level structural information into two neighborhoods. Yuan et al. [30] tried to achieve the feature alignment from multiple levels, i.e., image level, feature level, class level, and instance level, which can better address the unsupervised domain adaptation task.

However, the above methods that utilize class- or instance-level domain adaptation techniques have two major limitations, which may make them unavailable in HSI classification tasks. On the one hand, most of them reduce the domain shift to extract domain-invariant features by minimizing the statistical distance between similar samples of both domains in a latent space. An intuitive premise of these operations is that if target-domain samples are close to a certain class of source-domain clusters in a latent space, they should belong to the same class. Since adversarial learning may weaken the discriminability of features [31], the extracted features will contain a lot of nondiscriminative information. Samples with similar features may be classified as different classes. On the other hand, most of these methods strongly rely on the accuracy of predicted pseudo-labels. They utilize pseudo-labels directly without any processing. Unfortunately, pseudo-labels may not be completely reliable, and noisy pseudo-labels will lead to negative transfer between irrelevant classes.

To address the above issues, a soft instance-level domain adaptation with virtual classifier (SILDA-VC) is proposed for unsupervised HSI classification. The main contributions include the following.

1) We construct a feature similarity metric-based virtual classifier to output class probabilities of target-domain samples as auxiliary variables. By minimizing the real and virtual classifier determinacy disparity (RVCDD) loss, cross-domain samples with similar features are encouraged to be classified into the same class, thus enhancing the discriminability of the extracted high-level features.
2) To reduce the influence of noisy pseudo-labels, for each target-domain sample, the confidence coefficients are assigned to its corresponding positive and negative samples in the source domain. Also, we construct and minimize a soft prototype contrastive (SPC) loss. This will not only reduce the distance between similar samples in a latent space but also increase the predictive determinacy of samples in the target domain.
3) We perform comparative experiments with several existing domain adaptation methods and achieve the highest overall accuracy (OA) and kappa coefficient with low time consumption. Moreover, we define a classifier predictive consistency (CPC) loss to clearly demonstrate the predictive consistency between the real and virtual classifiers.

The structure of this article is organized as follows. The proposed SILDA-VC is elaborated in Section II. Experimental results and analysis are discussed in Section III, and the conclusion is drawn in Section IV.

II. SILDA-VC for HSI Classification

A. Framework of SILDA-VC

The framework of SILDA-VC is shown in Fig. 1, which includes four parts.

1) *Graph Convolutional Network (GCN)-Based Feature Extraction:* The spectral vectors and spatial coordinate information of source- and target-domain HSIs are utilized to construct spectral–spatial graphs of source and target domains, respectively, and high-level features $\boldsymbol{h}_i^{(\mathrm{S})}$ and $\boldsymbol{h}_i^{(\mathrm{T})}$ are extracted by jointly inputting the spectral vectors and the spectral–spatial graphs into the GCN.
2) *Reduction of Divergence Between Real and Virtual Classifiers:* The real classifier is trained through labeled source-domain features $\boldsymbol{h}_i^{(\mathrm{S})}$. The virtual classifier is constructed to output the feature similarity $\phi(\boldsymbol{h}_i^{(\mathrm{T})}, \boldsymbol{P}_c^{(\mathrm{S})})$ between target-domain samples $\boldsymbol{h}_i^{(\mathrm{T})}$ and class prototypes of source domain $\boldsymbol{P}_c^{(\mathrm{S})}$. The predictive consistency between the real and virtual classifiers is enhanced by minimizing the RVCDD loss.
3) *Soft Instance-Level Domain Adaptation:* To reduce the influence of noisy pseudo-labels when adapting two

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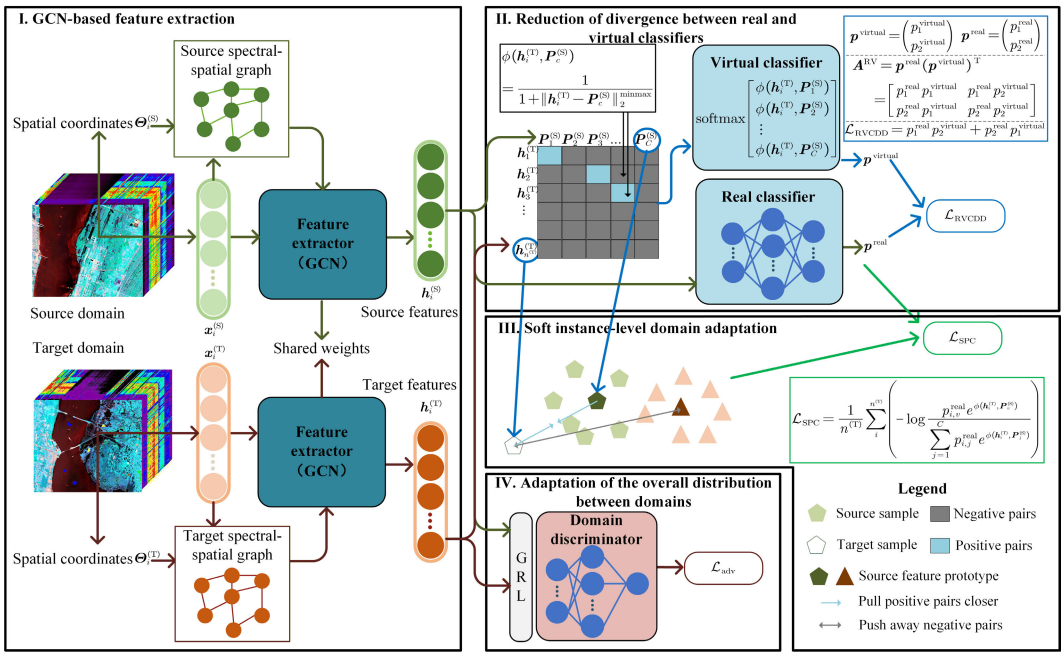


Fig. 1. Framework of SILDA-VC for HSI classification.

domains, we assign different confidence coefficients to each pair of positive and negative samples corresponding to the target-domain sample and adapt two domains in an instance-level way by minimizing the SPC loss.

4) *Adaptation of the Overall Distribution Between Domains:* The overall distribution divergence between domains is reduced through the adversarial training, in which a gradient reversal layer (GRL) added between the feature extractor and domain discriminator can make the parameter updating of adversarial training conduct in an end-to-end way.

B. GCN-Based Feature Extraction

Given a source-domain HSI $\boldsymbol{X}^{(\mathrm{S})} = [\boldsymbol{x}_1^{(\mathrm{S})}, \ldots, \boldsymbol{x}_{n^{(\mathrm{S})}}^{(\mathrm{S})}]^{\mathrm{T}}$ and its corresponding spatial coordinates $\boldsymbol{\Theta}_i^{(\mathrm{S})} = \{\Theta_{x,i}^{(\mathrm{S})}, \Theta_{y,i}^{(\mathrm{S})}\}$, where $\boldsymbol{X}^{(\mathrm{S})} \in \mathbb{R}^{n^{(\mathrm{S})} \times b}$, $\boldsymbol{\Theta}^{(\mathrm{S})} \in \mathbb{R}^{n^{(\mathrm{S})} \times 2}$, and $n^{(\mathrm{S})}$ is the number of source-domain samples, define an undirected graph $\boldsymbol{G}^{(\mathrm{S})} = \{\boldsymbol{V}^{(\mathrm{S})}, \boldsymbol{A}^{(\mathrm{S})}\}$, where $\boldsymbol{V}^{(\mathrm{S})}$ is the set of nodes $\{\boldsymbol{v}_1^{(\mathrm{S})}, \ldots \boldsymbol{v}_{n^{(\mathrm{S})}}^{(\mathrm{S})}\}$. Each node $\boldsymbol{v}_i^{(\mathrm{S})}$ represents a feature vector of a pixel, $\boldsymbol{v}_i^{(\mathrm{S})} = \boldsymbol{x}_i^{(\mathrm{S})}$. $\boldsymbol{A}^{(\mathrm{S})} \in \mathbb{R}^{n^{(\mathrm{S})} \times n^{(\mathrm{S})}}$ is a symmetric and sparse adjacency matrix whose elements $A_{i,j}^{(\mathrm{S})}$ is denoted as the weight of the edge connecting nodes $\boldsymbol{v}_i^{(\mathrm{S})}$ and $\boldsymbol{v}_j^{(\mathrm{S})}$. Then, $A_{i,j}^{(\mathrm{S})}$ can be calculated as

$$A_{i,j}^{(\mathrm{S})} = \begin{cases} e^{\frac{-l_{i,j}^{(\mathrm{S})}}{\sigma^2}}, & \text{if } \boldsymbol{v}_i^{(\mathrm{S})} \in \Xi\left(\boldsymbol{v}_j^{(\mathrm{S})}\right) \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\Xi(\boldsymbol{v}_j^{(\mathrm{S})})$ is the nearest neighbors set of $\boldsymbol{v}_j^{(\mathrm{S})}$ and σ is the bandwidth of Gaussian kernel function. $l_{i,j}^{(\mathrm{S})}$ is the spectral–spatial distance, which is used as the distance metric [32]

$$l_{i,j}^{(\mathrm{S})} = l_{i,j}^{(\mathrm{S}),\mathrm{spe}} + \psi l_{i,j}^{(\mathrm{S}),\mathrm{spa}} \tag{2}$$

where $l_{i,j}^{(\mathrm{S}),\mathrm{spe}} = \|\boldsymbol{v}_i^{(\mathrm{S})} - \boldsymbol{v}_j^{(\mathrm{S})}\|_2$ is the Euclidean distance between the spectral vectors of two nodes and $l_{i,j}^{(\mathrm{S}),\mathrm{spa}} = ((\Theta_{x,i}^{(\mathrm{S})} - \Theta_{x,j}^{(\mathrm{S})})^2 + (\Theta_{y,i}^{(\mathrm{S})} - \Theta_{y,j}^{(\mathrm{S})})^2)^{1/2}$ is the Euclidean distance between spatial coordinates of two nodes. ψ is the weighting factor used to control the relative importance of the spectral and spatial distances. If two nodes are more similar, $l_{i,j}^{(\mathrm{S})}$ is smaller and $A_{i,j}^{(\mathrm{S})}$ is larger. Compared with the spectral distance-based method, the GCN based on spectral–spatial graph can alleviate the effect of different ground objects with similar spectrum and the same ground objects with different spectrums in HSI. To avoid the numerical instability and gradient vanishing that may arise in the convolution operation, we normalize the adjacency matrix as follows:

$$\tilde{\boldsymbol{A}}^{(\mathrm{S})} = \tilde{\boldsymbol{D}}^{(\mathrm{S})-\frac{1}{2}}(\boldsymbol{A}^{(\mathrm{S})} + \boldsymbol{I})\tilde{\boldsymbol{D}}^{(\mathrm{S})-\frac{1}{2}} \tag{3}$$

where $\boldsymbol{I}$ is the identity matrix, $\boldsymbol{A}^{(\mathrm{S})} + \boldsymbol{I}$ is used to make the graph convolution operation retain information of its own nodes when aggregating the information of neighboring nodes, and $\tilde{\boldsymbol{D}}^{(\mathrm{S})}$ is the degree matrix corresponding to $\boldsymbol{A}^{(\mathrm{S})} + \boldsymbol{I}$. The sum of each row and each column of $\tilde{\boldsymbol{A}}^{(\mathrm{S})}$ obtained after the normalization operation is 1. Then, the features extracted by the $(m+1)$th layer of GCN can be expressed as

$$\boldsymbol{H}^{(\mathrm{S})m+1} = \phi(\tilde{\boldsymbol{A}}^{(\mathrm{S})}\boldsymbol{H}^{(\mathrm{S})m}\boldsymbol{W}^m) \tag{4}$$

where $\phi(\cdot)$ is the rectified linear unit (ReLU), $\boldsymbol{H}^{(\mathrm{S})m+1}$ and $\boldsymbol{H}^{(\mathrm{S})m}$ denote the output and input features of network in the $(m+1)$th layer, respectively, for the first layer $\boldsymbol{H}^{(\mathrm{S})0} = \boldsymbol{X}^{(\mathrm{S})}$, and $\boldsymbol{W}^m$ is the filter matrix in the mth layer. Finally, the high-level features of HSI can be extracted by GCN

$$\boldsymbol{h}_i^{(\mathrm{S})} = F_{\boldsymbol{\theta}_F}\left(\boldsymbol{x}_i^{(\mathrm{S})}\right) \tag{5}$$

where $F_{\boldsymbol{\theta}_F}(\cdot)$ is the GCN-based feature extractor and $\boldsymbol{\theta}_F$ is the network parameter. In the same way, the target-domain

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features $\boldsymbol{h}_i^{(\mathrm{T})}$ can also be obtained using the GCN-based feature extractor.

C. Reduction of Divergence Between Real and Virtual Classifiers

Intuitively, if the target-domain samples are close to source-domain cluster of a certain class in a latent space, they should belong to the same class. Since adversarial learning weakens the discriminability of features [31], the extracted high-level features will contain much other information that is unbeneficial for classification. This will result in the possibility that similar samples are classified into different classes, which largely aggravates the problem of negative transfer. Inspired by the biclassifier determinacy maximization (BCDM) [33], a virtual classifier is constructed by using the feature similarity between target-domain samples and source-domain class prototype, which can greatly encourage the source- and target-domain samples with similar features to be classified into the same class.

We use the source-domain class prototype to represent the center of a certain class of source-sample cluster, and the cth class prototype can be defined as

$$\boldsymbol{P}_c^{(\mathrm{S})} = \frac{1}{n_c^{(\mathrm{S})}} \sum_{i=1}^{n_c^{(\mathrm{SS})}} F_{\boldsymbol{\theta}_F}\left(\boldsymbol{x}_{c,i}^{(\mathrm{S})}\right) \tag{6}$$

where $\boldsymbol{x}_{c,i}^{(\mathrm{S})}$ is the cth class source-domain sample and $n_c^{(\mathrm{S})}$ is the number of the cth class source-domain samples.

The constructed virtual classifier is not a real neural network, and it outputs the feature similarity between target-domain samples and class prototypes of source domain. A similarity matrix $\boldsymbol{A}^{\mathrm{sim}} \in \mathbb{R}^{n^{(\mathrm{T})} \times n^C}$ is first calculated, where $n^{(\mathrm{T})}$ is the number of samples in target domain and n^C is the number of classes. $A_{i,j}^{\mathrm{sim}}$ is the similarity between the ith target-domain sample and the jth source-domain class prototype. To eliminate the amplitude difference of features among various dimensions, a revised normalized inverse Euclidean distance is utilized to calculate the feature similarity between target-domain samples and source-domain class prototype

$$\phi\left(\boldsymbol{h}_i^{(\mathrm{T})}, \boldsymbol{P}_c^{(\mathrm{S})}\right) = \frac{1}{1 + \left\| \boldsymbol{h}_i^{(\mathrm{T})} - \boldsymbol{P}_c^{(\mathrm{S})} \right\|_2^{\min\max}} \tag{7}$$

where $\|\cdot\|_2^{\min\max}$ is the min-max normalized Euclidean distance. Here, a softmax function is added to the output of virtual classifier so that the obtained vectors can simultaneously represent: 1) the similarity between target-domain samples and source-domain class prototypes and 2) the predicted possibility of each target-domain sample belonging to each class. Then, the output of virtual classifier is

$$\boldsymbol{p}_i^{\mathrm{virtual}} = \mathrm{softmax}\left[A_{i,1}^{\mathrm{sim}}, A_{i,2}^{\mathrm{sim}}, \ldots, A_{i,n^C}^{\mathrm{sim}}\right]^{\mathrm{T}} \tag{8}$$

where $A_{i,n^C}^{\mathrm{sim}} = \phi\left(\boldsymbol{h}_i^{(\mathrm{T})}, \boldsymbol{P}_{n^C}^{(\mathrm{S})}\right)$ and $\boldsymbol{p}_i^{\mathrm{virtual}} \in \mathbb{R}^{n^C}$ denotes the class prediction of the virtual classifier for the ith target-domain sample.

The class prediction $\boldsymbol{p}_i^{\mathrm{real}}$ of the real classifier for the ith target-domain sample is denoted as

$$\boldsymbol{p}_i^{\mathrm{real}} = C_{\boldsymbol{\theta}_C}\left(F_{\boldsymbol{\theta}_F}\left(\boldsymbol{x}_i^{(\mathrm{T})}\right)\right) \tag{9}$$

where $C_{\boldsymbol{\theta}_C}(\cdot)$ is the real classifier, $\boldsymbol{\theta}_C$ is the network parameter, and $\boldsymbol{p}_i^{\mathrm{real}} \in \mathbb{R}^{n^C}$ is the softmax probabilistic output. To investigate the divergence between the real and virtual classifiers, the prediction relevance matrix is constructed as

$$\begin{aligned} \boldsymbol{A}_i^{\mathrm{RV}} &= \boldsymbol{p}_i^{\mathrm{real}}\left(\boldsymbol{p}_i^{\mathrm{virtual}}\right)^{\mathrm{T}} \\ &= \begin{bmatrix} p_{i,1}^{\mathrm{real}} p_{i,1}^{\mathrm{virtual}} & \cdots & p_{i,1}^{\mathrm{real}} p_{i,n^C}^{\mathrm{virtual}} \\ \vdots & \ddots & \vdots \\ p_{i,n^C}^{\mathrm{real}} p_{i,1}^{\mathrm{virtual}} & \cdots & p_{i,n^C}^{\mathrm{real}} p_{i,n^C}^{\mathrm{virtual}} \end{bmatrix} \end{aligned} \tag{10}$$

where $\boldsymbol{A}_i^{\mathrm{RV}} \in \mathbb{R}^{n^C \times n^C}$ is obtained by the product of predicted possibility from the real and virtual classifiers and $p_{i,c}^{\mathrm{real}}$ and $p_{i,c}^{\mathrm{virtual}}$ are the probabilities that the real and virtual classifiers assign class c to the ith target-domain sample, respectively. Therefore, $\boldsymbol{A}^{\mathrm{RV}}$ can effectively represent the relationship between similarity information from the virtual classifier and predicted possibility from the real classifier. Then, to enhance the predictive consistency between the real and virtual classifiers, we need to maximize the elements on the diagonal of $\boldsymbol{A}^{\mathrm{RV}}$ and minimize the remaining elements. Define RVCDD loss as

$$\begin{aligned} \mathcal{L}_{\mathrm{RVCDD}} &= \frac{1}{n^{(\mathrm{T})}} \sum_{i=1}^{n^{(\mathrm{T})}} \left(\sum_{m,n=1}^{n^C} A_{i,mn}^{\mathrm{RV}} - \sum_{m=1}^{n^C} A_{i,mm}^{\mathrm{RV}} \right) \\ &= \frac{1}{n^{(\mathrm{T})}} \sum_{i=1}^{n^{(\mathrm{T})}} \left(\sum_{m \neq n}^{n^C} A_{i,mn}^{\mathrm{RV}} \right) \end{aligned} \tag{11}$$

where $A_{i,mn}^{\mathrm{RV}}$ denotes the element in the mth row and the nth column of $\boldsymbol{A}_i^{\mathrm{RV}}$. $n^{(\mathrm{T})}$ is the number of target-domain samples, $\sum_{m,n=1}^{n^C} A_{i,mn}^{\mathrm{RV}} = 1$. $\mathcal{L}_{\mathrm{RVCDD}}$ contains all prediction probabilities of inconsistency between the real and virtual classifiers. Therefore, minimizing RVCDD loss can make the outputs of the real and virtual classifiers to be consistent, allowing the extracted high-level features to contain more information that is beneficial for classification.

D. Soft Instance-Level Domain Adaptation

Most of the class- and instance-level domain adaptation methods utilize pseudo-labels without any processing to reduce the domain shift finely by minimizing the statistical distance of both domains. These kinds of methods rely on the accuracy of pseudo-labels strongly. Unfortunately, pseudo-labels cannot be completely reliable. To reduce the influence of noisy pseudo-labels, we propose a soft instance-level domain adaptation, which uses the similarity matrix mentioned above to calculate the SPC loss. The source-domain class prototype with the highest similarity to the target-domain sample is taken as the positive sample $P_v^{(\mathrm{S})}$, where $v = \arg\max_v [\boldsymbol{p}_i^{\mathrm{virtual}}]_v$ and $[\cdot]_v$ denotes the vth element. The rest of the source-domain class prototypes are taken as negative samples. To further reduce the influence of noisy pseudo-labels, for each target-domain sample, we assign the confidence coefficients to its

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corresponding positive and negative samples in the source domain. In specific, the output of real classifier $\boldsymbol{p}_i^{\text{real}}$ is used as the confidence coefficient to evaluate the similarity of the positive and negative sample pairs. Then, the SPC loss is expressed as

$$\mathcal{L}_{\text{SPC}} = \frac{1}{n^{(\text{T})}} \sum_{i=1}^{n^{(\text{T})}} \left(-\log \frac{p_{i,v}^{\text{real}} e^{\phi(\boldsymbol{h}_i^{(\text{T})}, \boldsymbol{P}_v^{(\text{S})})}}{\sum_{j=1}^{n^C} p_{i,j}^{\text{real}} e^{\phi(\boldsymbol{h}_i^{(\text{T})}, \boldsymbol{P}_j^{(\text{S})})}} \right) \tag{12}$$

where $\phi(\boldsymbol{h}_i^{(\text{T})}, \boldsymbol{P}_v^{(\text{S})})$ is the similarity between the target-domain sample and positive sample and $p_{i,v}^{\text{real}}$ is the probability that the real classifier assigns target-domain samples to the vth class. In this article, we use $p_{i,v}^{\text{real}}$ as the confidence coefficient for $\boldsymbol{P}_v^{(\text{S})}$ to be the positive sample of $\boldsymbol{h}_i^{(\text{T})}$. Minimizing $\mathcal{L}_{\text{SPC}}$ can not only decrease the distance between target-domain samples and positive samples but also increase the distance between target-domain samples and negative samples. In addition, with the help of confidence coefficient, the determinacy of classifier can be enhanced.

Compared with general instance-level domain adaptation methods, SILDA-VC has the following advantages: 1) SILDA-VC can adapt target-domain samples to the cluster center of source domain, which can largely reduce the risk of adapting target-domain samples to outlier source-domain samples, and 2) SILDA-VC can assign different confidence coefficients to each pair of positive and negative samples corresponding to the target-domain sample, which can greatly alleviate the effect of noisy pseudo-labels on the classification performance.

E. Adaptation of Overall Distribution Between Domains

The domain adversarial strategy aims to reduce the overall distribution divergence between the source and target domains, which can be realized by performing a two-player game between a feature extractor and a domain discriminator. The domain discriminator tries to distinguish between the source- and target-domain features, while the feature extractor tries to fool the domain discriminator. Meanwhile, the feature extractor and classier are trained simultaneously to achieve low classification error on source samples by reducing the empirical risk of source domain. In summary, the objective function of domain adversarial loss can be defined as

$$\begin{aligned}\mathcal{L}_{\text{adv}} = &\frac{1}{n^{(\text{S})}} \sum_{i=1}^{n^{(\text{S})}} \sum_{c=1}^{n^C} \mathcal{L}_{\text{ce}}\left(C_{\boldsymbol{\theta}_C}\left(F_{\boldsymbol{\theta}_F}\left(\boldsymbol{x}_i^{(\text{S})}\right)\right), y_{i,c}^{(\text{S})}\right) \\ &- \frac{\epsilon}{n^{(\text{S})} + n^{(\text{T})}} \sum_{\boldsymbol{x}_i \in \left(\mathcal{D}^{(\text{S})} \cup \mathcal{D}^{(\text{T})}\right)} \mathcal{L}_{\text{bce}}\left(D_{\boldsymbol{\theta}_D}\left(F_{\boldsymbol{\theta}_F}(\boldsymbol{x}_i)\right), d_i\right)\end{aligned} \tag{13}$$

where $\mathcal{L}_{\text{ce}}(\cdot,\cdot)$ is the cross-entropy loss, $y_{i,c}^{(\text{S})}$ is the class label of the ith source-domain sample, $\mathcal{D}^{(\text{S})}$ and $\mathcal{D}^{(\text{T})}$ denote the source domain with $n^{(\text{S})}$ samples and target domain with $n^{(\text{T})}$ samples, respectively, $\mathcal{L}_{\text{bce}}(\cdot,\cdot)$ is the binary cross-entropy loss, d_i is the corresponding domain label of sample x_i, $D_{\boldsymbol{\theta}_D}(\cdot)$ is the domain discriminator, and $\boldsymbol{\theta}_D$ is the network parameter. $\epsilon = (1 - e^{\alpha\rho})/(1 + e^{\alpha\rho})$ denotes the tradeoff parameter, where α determines the changing speed of ϵ and $\rho \in [0, 1]$ is the ratio of current to maximum iterations. To implement the domain adversarial strategy, a GRL is added between the feature extractor and domain discriminator, which can make the parameter updating of adversarial training conduct in an end-to-end way.

F. Overall Objective Function of SILDA-VC

The overall objective function of SILDA-VC consists of three components: domain adversarial loss $\mathcal{L}_{\text{adv}}$, RVCDD $\mathcal{L}_{\text{RVCDD}}$, and SPC loss $\mathcal{L}_{\text{SPC}}$

$$J(\boldsymbol{\theta}_F, \boldsymbol{\theta}_C, \boldsymbol{\theta}_D) = \mathcal{L}_{\text{adv}} + \epsilon_{\text{RV}} \mathcal{L}_{\text{RVCDD}} + \epsilon_{\text{SPC}} \mathcal{L}_{\text{SPC}} \tag{14}$$

where ϵ_{RV} and ϵ_{SPC} are the tradeoff parameters to balance the contributions of each loss and $\boldsymbol{\theta}_F$, $\boldsymbol{\theta}_C$, and $\boldsymbol{\theta}_D$ are the network parameters of the feature extractor, real classifier, and domain discriminator, respectively.

III. Experiments and Analysis

To verify the validity of SILDA-VC, we organize experiments and analyses on four HSI datasets, which are Botswana (BOT), Kennedy Space Center (KSC), Pavia Center (PC), Pavia University (PU), and HyRANK. All experiments are performed on a computer equipped with 3.80-GHz Intel Core I7-10700KF CPU 32 GB of RAM, and RTX 2080Ti GPU, and the experimental environment is Pytorch. We use three indicators to evaluate the experimental results, which are OA, kappa coefficient, and time consumption. In addition, all experiments are repeated ten times to eliminate the influence of random factors on the test results.

A. Dataset Description

The BOT dataset consists of three HSIs that were acquired by NASA EO-1 satellite at Okavango Delta, BOT in May (BOT5), June (BOT6), and July (BOT7) 2001 with Hyperion sensor at 7.7 km in 242 bands. After removing the water absorption and low signal-to-noise ratio bands, the remaining 145 bands are used as candidate features. BOT5, BOT6, and BOT7 all contain nine recognized classes, consisting of 1476 × 256 pixels, which contain 3014, 2871, and 4997 labeled samples, respectively.

The KSC dataset consists of two HSIs obtained by NASA AVIRIS in different areas over KSC, Florida, on March 23, 1996, in 224 bands at an altitude of 20 km. After removing the water absorption and low signal-to-noise ratio bands, the remaining 176 bands are used as candidate features. The KSC and KSC3 contain ten recognized classes, consisting of 512 × 614 pixels, which contain 3784 and 3854 labeled samples, respectively.

The PC and PU datasets each contain an HSI, which was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor in Pavia, Italy. Both PC and PU contain 102 bands (the last band of PU is removed). In this article, the central regions of PC and PU are selected for experiments. The PC and PU contain four recognized classes, consisting of 601 × 251 and 320 × 180 pixels, which contain 5208 and 5094 labeled samples, respectively.

The HyRANK dataset consists of two HSIs, which are Dioni and Loukia. The HyRANK dataset has been developed in

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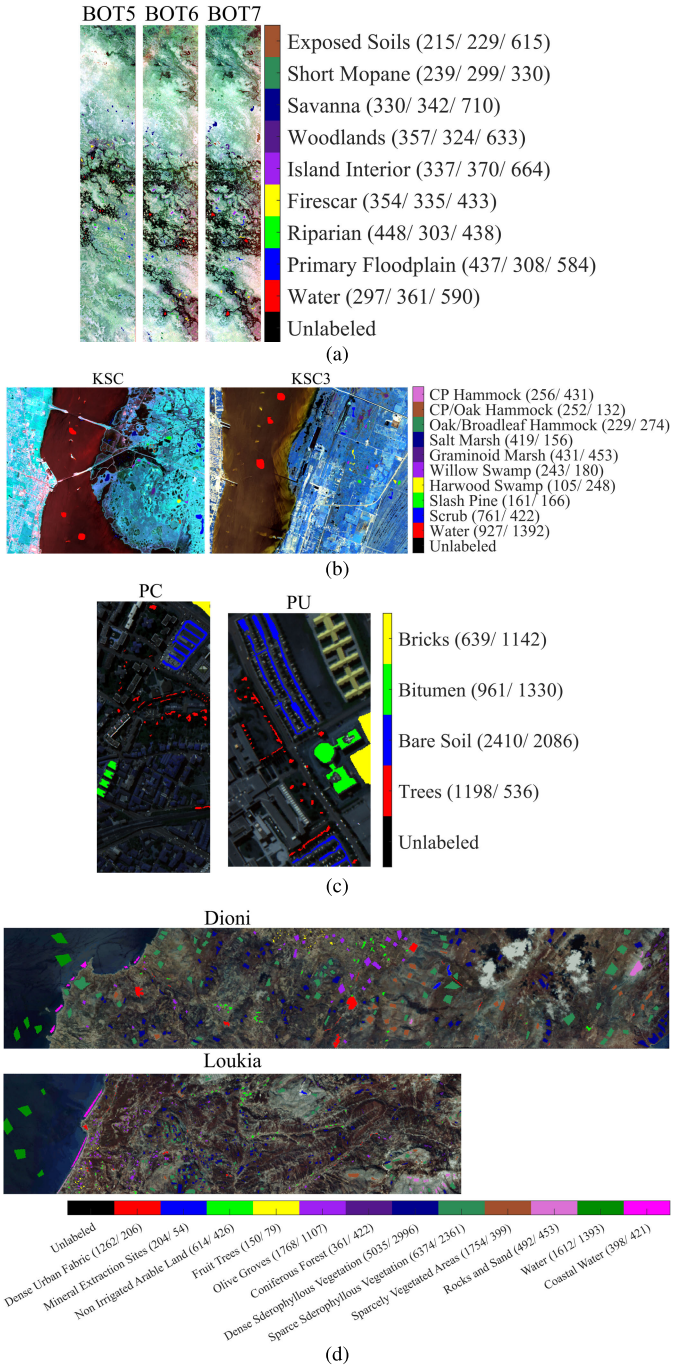


Fig. 2. False-color images of HSIs. (a) BOT. (b) KSC. (c) PC and PU. (d) HyRANK.

the framework of the International Society for Photogrammetry and Remote Sensing Scientific Initiatives and was acquired by the NASA EO-1 satellite. Both Dioni and Loukia contain 176 bands and 12 recognized classes, consisting of 250 × 1376 and 249 × 945 pixels, which contain 20 024 and 10 317 labeled samples, respectively. Compared with BOT, KSC, PC, and PU, the HyRANK dataset has more complicated spatial land-cover distribution, most land-cover categories, and similar spectral distribution, which is a more challenging dataset for cross-scene HSI classification methods [34].

False-color images and more sample information are shown in Fig. 2, and 12 transfer tasks are constructed for experiments: BOT5→BOT6, BOT6→BOT5, BOT5→BOT7, BOT7→BOT5, BOT6→BOT7, BOT7→BOT6, KSC→KSC3, KSC3→KSC, PC→PU, PU→PC, Dioni→Loukia, and Loukia→Dioni.

B. Parameter Settings

SILDA-VC consists of a feature extractor, a classifier, and a domain discriminator. For the feature extractor, according to [35], a single-layer GCN cannot achieve the effective information aggregation of intraclass samples, while GCN with too many layers will lead to oversmoothing and overaggregation of interclass samples. Therefore, a three-layer GCN based on spectral–spatial graph is used for all tasks. The ReLU activation function, dropout strategy, and batch normalization are used after each hidden layer. For the domain discriminator, a multilayer perceptron (MLP) with single hidden layer is selected for all tasks. The ReLU activation function is used after the hidden layer, and the sigmoid activation function is used in the output layer. For the classifier, an MLP with no hidden layer is selected for all tasks.

Too few hidden-layer nodes will lead to insufficient fitting ability, while too many hidden-layer nodes will not only lead to the existence of redundant nodes but also increase the risk of overfitting. Therefore, the number of nodes is set as 350 per hidden layer for feature extractor for all tasks, and 100 nodes per hidden layer are selected for the domain discriminator for all tasks by errors and trials.

According to [36], ψ is set to 10. The bandwidth σ of the Gaussian kernel function is set to 20. For BOT5→BOT6, BOT6→BOT5, BOT7→BOT6, Dioni→Loukia, and Loukia→Dioni, the number of nearest neighbors k is set to 8, and for the rest tasks, k is set to 20. The optimizer chosen for the network training is Adam. The learning rate is set to $\eta = \eta_0/(1 + 10\rho)^i$, where i is set to 0.75. For BOT5→BOT6, BOT6→BOT5, BOT7→BOT6, PC→PU, PU→PC, Dioni→Loukia, and Loukia→Dioni, the initial learning rate η_0 is set to 0.0005, and for the rest of the tasks, η_0 is set to 0.00005. In the domain adversarial loss $\mathcal{L}_{\text{adv}}$, α in the definition of ϵ is set to 10. In the overall objective function, ϵ_{SPC} and ϵ_{RV} are set to 0.0005 and 10 for BOT5→BOT6, BOT6→BOT5, BOT6→BOT7, and BOT7→BOT6, to 0.0005 and 100 on Dioni→Loukia and Loukia→Dioni, and to 2 and 100 for the rest of tasks. For the total number of training iterations, all tasks are set to 1000 except PC→PU and PU→PC, which are set to 200.

C. Comparative Experiments

In order to verify the effectiveness of SILDA-VC, ten benchmark methods are selected to conduct the comparative experiments: 1) two deep learning methods (DNN and GCN); 2) eight deep transfer networks (DANN [18], deep correlation alignment (D-CORAL) [37], deep adaptation network (DAN) [38], MADA [25], joint CORAL-based graph neural network (JCGNN) [34], CDA [24], BCDM [33], and adaptive graph adversarial network (AGAN) [39]). In AGAN, a spectral–spatial graph-based GCN is used as the feature extractor. DNN and GCN are the versions of DANN and

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TABLE I
COMPARISON OF OA (%)

Task	DNN	GCN	DANN	D-CORAL	DAN	MADA	JCGNN	CDA	BCDM	AGAN	SILDA-VC
BOT5→BOT6	82.38	90.39	87.95	87.63	87.41	89.27	92.71	90.25	91.12	93.42	**94.09**
BOT6→BOT5	81.29	88.17	88.32	86.74	86.46	88.33	92.90	92.57	91.24	93.79	**97.48**
BOT5→BOT7	67.80	72.12	78.21	74.09	73.58	76.33	78.65	78.29	74.34	78.87	**79.62**
BOT7→BOT5	78.37	81.02	82.16	80.92	80.97	82.18	77.72	82.58	88.65	86.54	**89.99**
BOT6→BOT7	83.37	83.43	84.34	82.89	84.11	85.03	82.85	83.53	80.99	84.33	**85.27**
BOT7→BOT6	90.60	91.54	89.26	88.14	88.85	91.63	88.44	89.24	94.08	86.39	**95.26**
KSC→KSC3	68.60	71.54	70.61	71.14	70.02	70.71	73.00	73.46	69.80	73.02	**76.93**
KSC3→KSC	61.95	68.03	68.43	69.31	69.90	72.10	73.41	68.55	70.24	70.75	**75.67**
PC→PU	89.87	88.48	91.08	90.97	85.96	91.44	92.93	89.75	90.11	91.46	**94.01**
PU→PC	80.32	89.25	83.03	83.54	77.70	83.21	89.41	83.27	84.04	88.92	**92.54**
Dioni→Loukia	68.15	68.32	68.88	69.21	70.23	72.56	67.23	73.44	73.10	67.49	**73.46**
Loukia→Dioni	67.54	64.04	66.58	65.46	67.49	68.07	62.22	69.02	65.99	63.31	**70.01**

TABLE II
COMPARISON OF KAPPA COEFFICIENT

Task	DNN	GCN	DANN	D-CORAL	DAN	MADA	JCGNN	CDA	BCDM	AGAN	SILDA-VC
BOT5→BOT6	0.8012	0.8915	0.8642	0.8605	0.8580	0.8791	0.9178	0.8893	0.8999	0.9258	**0.9334**
BOT6→BOT5	0.7887	0.8664	0.8681	0.8502	0.8471	0.8682	0.9197	0.9160	0.9010	0.9277	**0.9716**
BOT5→BOT7	0.6000	0.6839	0.7543	0.7075	0.7016	0.7328	0.7593	0.7553	0.7104	0.7604	**0.7687**
BOT7→BOT5	0.7557	0.7862	0.7986	0.7846	0.7854	0.7989	0.7487	0.8035	0.8719	0.8479	**0.8865**
BOT6→BOT7	0.8123	0.8126	0.8235	0.8070	0.8204	0.8313	0.8068	0.8144	0.7862	0.8231	**0.8337**
BOT7→BOT6	0.8939	0.9045	0.8790	0.8662	0.8743	0.9056	0.8695	0.8788	0.9332	0.8467	**0.9466**
KSC→KSC3	0.6147	0.6491	0.6412	0.6462	0.6327	0.6421	0.6679	0.6750	0.6311	0.6673	**0.7166**
KSC3→KSC	0.5649	0.6306	0.6330	0.6413	0.6449	0.6749	0.6913	0.6297	0.6507	0.6632	**0.7167**
PC→PU	0.8586	0.8385	0.8751	0.8735	0.8040	0.8800	0.9009	0.8567	0.8618	0.8792	**0.9138**
PU→PC	0.7095	0.8430	0.7547	0.7631	0.6774	0.7565	0.8426	0.7582	0.7737	0.8421	**0.8905**
Dioni→Loukia	0.6161	0.6147	0.6218	0.6270	0.6360	0.6675	0.6031	0.6774	0.6737	0.6024	**0.6756**
Loukia→Dioni	0.6042	0.5465	0.5893	0.5765	0.6019	0.6066	0.5426	0.6176	0.5902	0.5530	**0.6308**

TABLE III
COMPARISON OF CONSUMED TIME (S)

Task	DNN	GCN	DANN	D-CORAL	DAN	MADA	JCGNN	CDA	BCDM	AGAN	SILDA-VC
BOT5→BOT6	478.89	27.27	550.49	423.93	122.59	757.48	206.95	1677.70	1466.23	46.34	63.42
BOT6→BOT5	476.25	27.03	545.45	427.28	120.26	738.80	210.45	1631.19	1389.09	51.91	58.09
BOT5→BOT7	765.88	29.41	892.41	704.92	124.30	1222.78	222.96	2866.15	2316.22	60.42	67.18
BOT7→BOT5	767.03	30.64	885.80	637.78	120.56	1182.21	217.72	2890.24	2286.98	58.81	65.22
BOT6→BOT7	758.11	29.35	887.05	688.29	124.86	1225.80	221.56	2662.44	2287.10	59.89	68.64
BOT7→BOT6	764.03	30.64	884.53	624.34	119.94	1160.20	221.73	2627.21	2313.07	55.45	67.38
KSC→KSC3	633.58	29.32	730.26	607.91	139.40	965.38	208.28	2559.81	1865.94	62.32	68.80
KSC3→KSC	634.32	29.58	720.06	603.94	137.67	960.10	151.94	2436.14	1819.47	58.03	75.46
PC→PU	164.24	10.50	180.47	131.04	99.61	185.88	44.16	630.24	500.47	15.13	16.67
PU→PC	163.81	10.63	180.37	130.58	99.67	184.92	43.93	655.59	495.14	14.91	16.48
Dioni→Loukia	1923.99	102.23	2175.51	2270.63	161.37	5457.80	214.88	12030.79	6740.57	137.94	156.44
Loukia→Dioni	1888.03	128.37	2249.56	2320.09	187.06	5819.23	346.73	11241.24	6809.11	140.21	156.50

AGAN after removing the domain discriminator, respectively. Classification performances obtained by different methods are shown in Tables I–III. It can be seen that the following conditions hold.

1) DANN and AGAN perform better than DNN and GCN on most tasks. This is because DNN and GCN will overfit when only the source-domain samples are used in the training process. Since the distributions of source and target domains own differences, the network trained upon source-domain samples will lead to serious misclassification on the target-domain testing samples.
2) Compared with DANN, AGAN achieves higher OAs on most tasks. This is because the spectral–spatial graph-based GCN is used as a feature extractor in AGAN, which can make full use of the spectral and spatial information of HSI.
3) Compared with domain-level deep transfer networks (DANN, D-CORAL, DAN, and BCDM), class-level deep transfer networks (MADA, JCGNN, and CDA)

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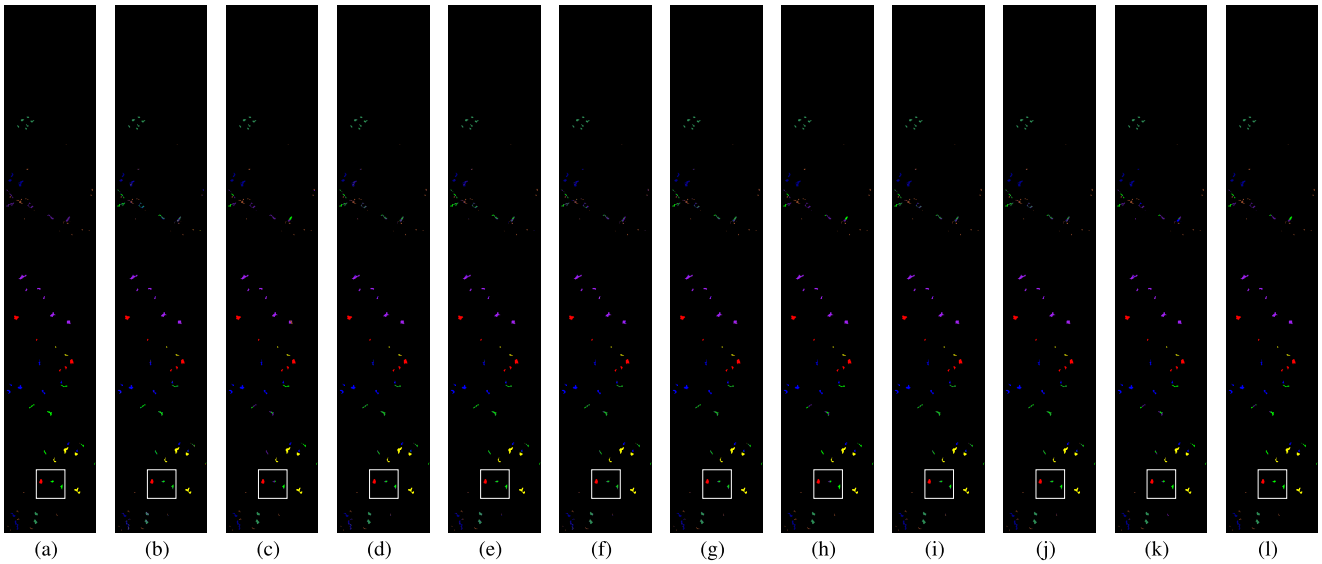


Fig. 3. Classification maps obtained by different methods on BOT5→BOT6. (a) Ground truth. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

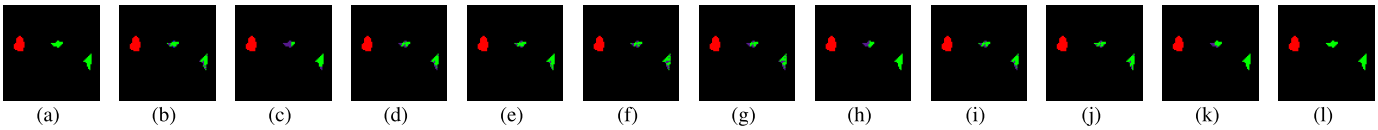


Fig. 4. Classification maps of a local region obtained by different methods on BOT5→BOT6. (a) Ground truth. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

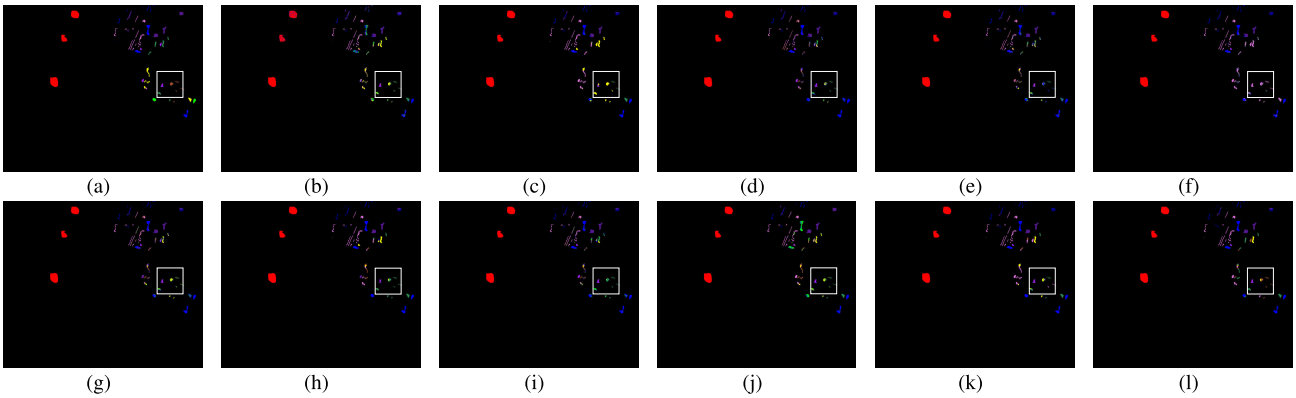


Fig. 5. Classification maps obtained by different methods on KSC→KSC3. (a) Ground truth. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

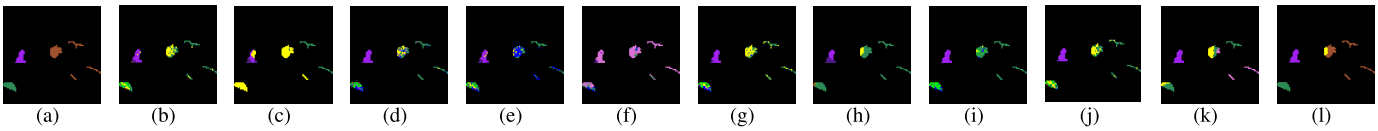


Fig. 6. Classification maps of a local region obtained by different methods on KSC→KSC3. (a) Ground truth. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

can achieve better classification results on most tasks. This is because the multimodal structure of data can be considered in the class-level adaptation methods.

4) SILDA-VC achieves the highest OA and kappa coefficient on all tasks, and the reasons are twofold. First, the predictive consistency between the real and virtual classifiers is enhanced by reducing the divergence between two classifiers, allowing the hidden-layer features to contain more information that is beneficial for classification and enhancing the discriminability of the features. Second, SILDA-VC can adapt the source- and target-domain samples more finely and increase the determinacy of the classifier by minimizing the SPC loss.

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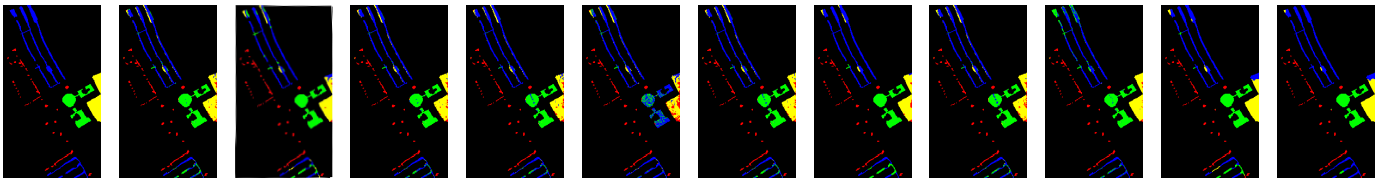


Fig. 7. Classification maps obtained by different methods on PC→PU. (a) Ground truth. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

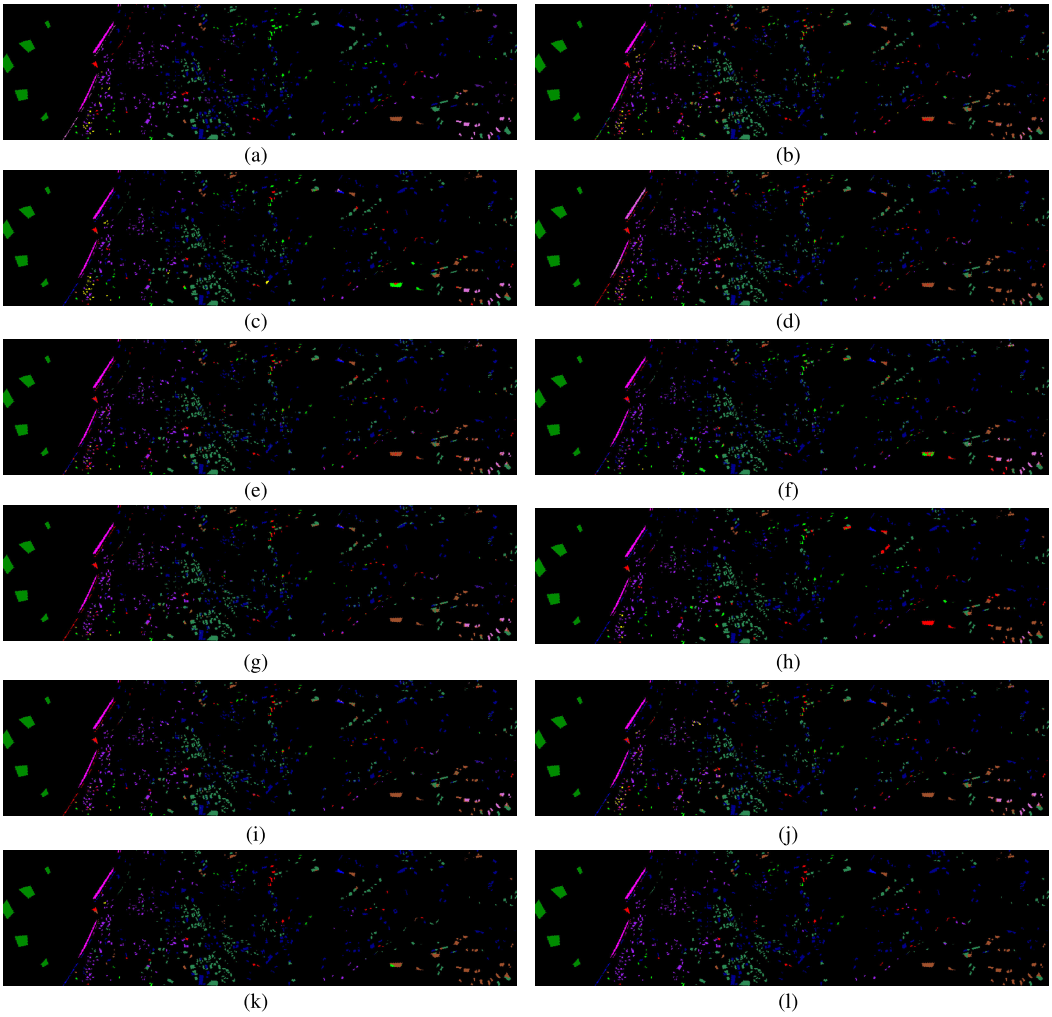


Fig. 8. Classification maps obtained by different methods on Dioni→Loukia. (a) Ground truth. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

5) The consumed time of SILDA-VC is neither the minimum nor maximum, which is in an acceptable range. BCDM, MADA, and CDA consume much time than other methods. The reason is that BCDM needs to update the network parameters several times in each iteration, while MADA and CDA need extra training cost of multiple domain discriminators.

To further verify the effectiveness of SILDA-VC, classification results of different methods on target domain are visualized. Taking BOT5→BOT6, KSC→KSC3, PC→PU, and Dioni→Loukia as examples, classification maps obtained by different methods are shown in Figs. 3–8. Since samples of BOT and KSC datasets are scattered on the maps, which makes it not easy to be observed, local regions are cropped and observed in Figs. 4 and 6. It can be seen that classification maps obtained by SILDA-VC are smoother and most similar to the ground truth. Besides, SILDA-VC can classify some samples that are difficult to be correctly classified, such as CP/Oak Hammock (brown) in the local region of KSC3.

To verify the edge-preserving ability of different methods, we give the full classification maps of all comparative methods. Taking Dioni→Loukia and Loukia→Dioni as examples, it can be seen from Figs. 9 and 10 that the full classification maps obtained by SILDA-VC are cleaner, smoother,

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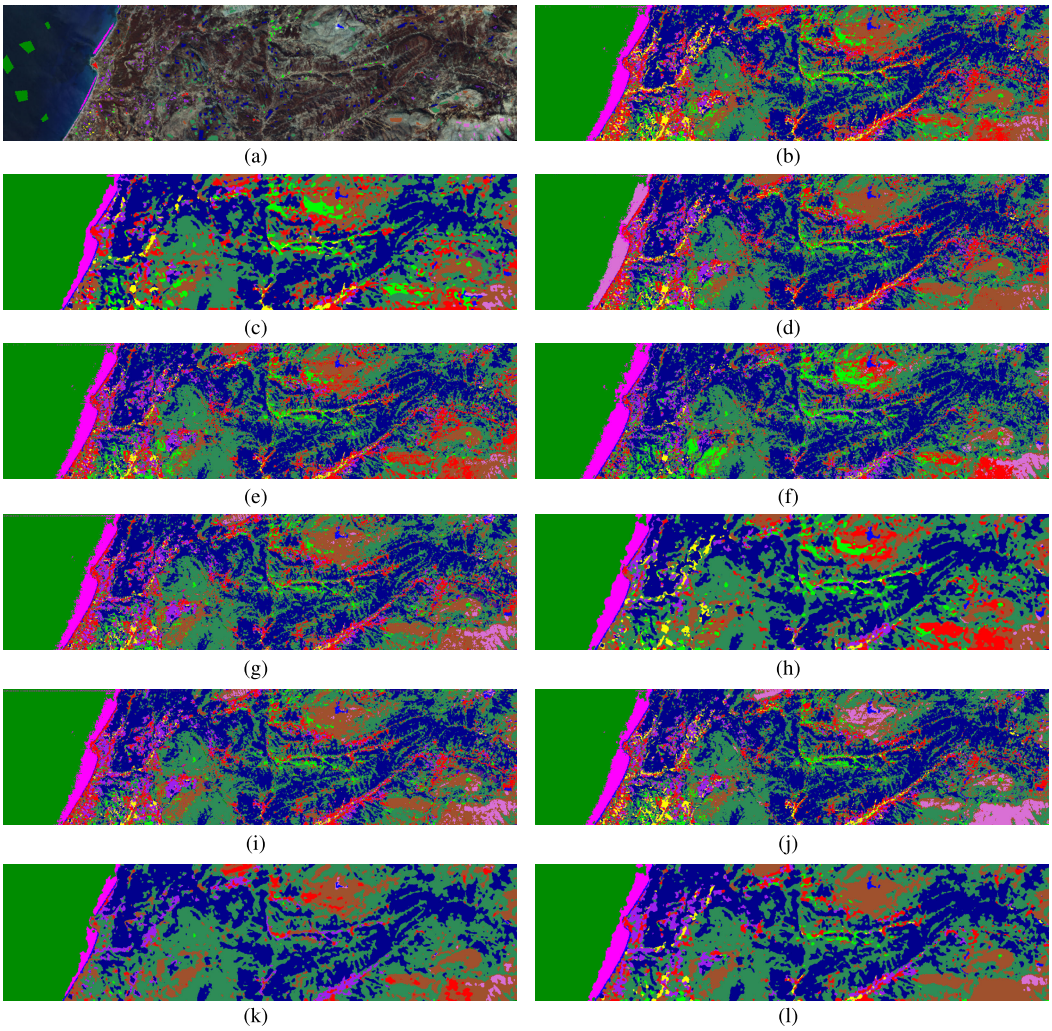


Fig. 9. Full classification maps obtained by different methods on Dioni→Loukia. (a) False-color image of Loukia. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

and abundant in detail, showing better edge-preserving ability. More specifically, taking the task Dioni→Loukia as an example, SILDA-VC can clearly separate Water (dark green) and Coastal Water (dark pink). However, the comparative methods more or less misclassify a few "Water" as "Coastal Water."

Moreover, we conduct the ablation experiment to demonstrate the importance of RVCDD loss and SPC loss, and the experimental results are listed in Table IV, where SILDA-VC (w/o SPC) means SILDA-VC without minimizing SPC loss and SILDA-VC (w/o RVCDD) means SILDA-VC without minimizing RVCDD loss. In the ablation experiment, SILDA-VC achieves higher OA compared with SILDA-VC (w/o SPC) and SILDA-VC (w/o RVCDD). This is because: 1) compared with SILDA-VC (w/o SPC), SILDA-VC is able to adapt the source- and target-domain samples more finely from the perspective of instance level, and besides, SILDA-VC can increase the determinacy of classifier and keep samples away from the decision boundary and 2) compared with SILDA-VC (w/o RVCDD), SILDA-VC enhances the discriminability of features by minimizing the RVCDD loss, which is beneficial for classification.

TABLE IV
OA (%) RESULT OF ABLATION STUDY

Task	AGAN	SILDA-VC (w/o SPC)	SILDA-VC (w/o RVCDD)	SILDA-VC
BOT5→BOT6	93.42	93.98	93.77	**94.09**
BOT6→BOT5	93.79	96.45	93.36	**97.48**
BOT5→BOT7	78.87	79.55	74.61	**79.62**
BOT7→BOT5	86.54	88.52	89.25	**89.99**
BOT6→BOT7	84.33	85.27	84.97	**85.27**
BOT7→BOT6	86.39	95.13	86.48	**95.26**
KSC→KSC3	73.02	74.60	74.81	**76.93**
KSC3→KSC	70.75	74.87	74.22	**75.67**
PC→PU	91.46	92.64	93.95	**94.01**
PU→PC	88.92	92.44	88.92	**92.54**
Dioni→Loukia	67.49	73.07	68.48	**73.46**
Loukia→Dioni	63.31	66.76	63.28	**70.01**

To demonstrate that SILDA-VC can enhance the predictive consistency between the real and virtual classifiers, we define a CPC loss: $\mathcal{L}_{\mathrm{CPC}} = (1/(n^{(\mathrm{T})})) \sum_{i=1}^{n^{(\mathrm{T})}} (\tilde{y}_{\mathrm{i}}^{\mathrm{real}} \bigoplus \tilde{y}_{\mathrm{i}}^{\mathrm{virtual}})$, where $\tilde{y}_{\mathrm{i}}^{\mathrm{real}}$ and $\tilde{y}_{\mathrm{i}}^{\mathrm{virtual}}$ denote the labels of the ith target-domain sample predicted by the real and virtual classifiers, respectively.

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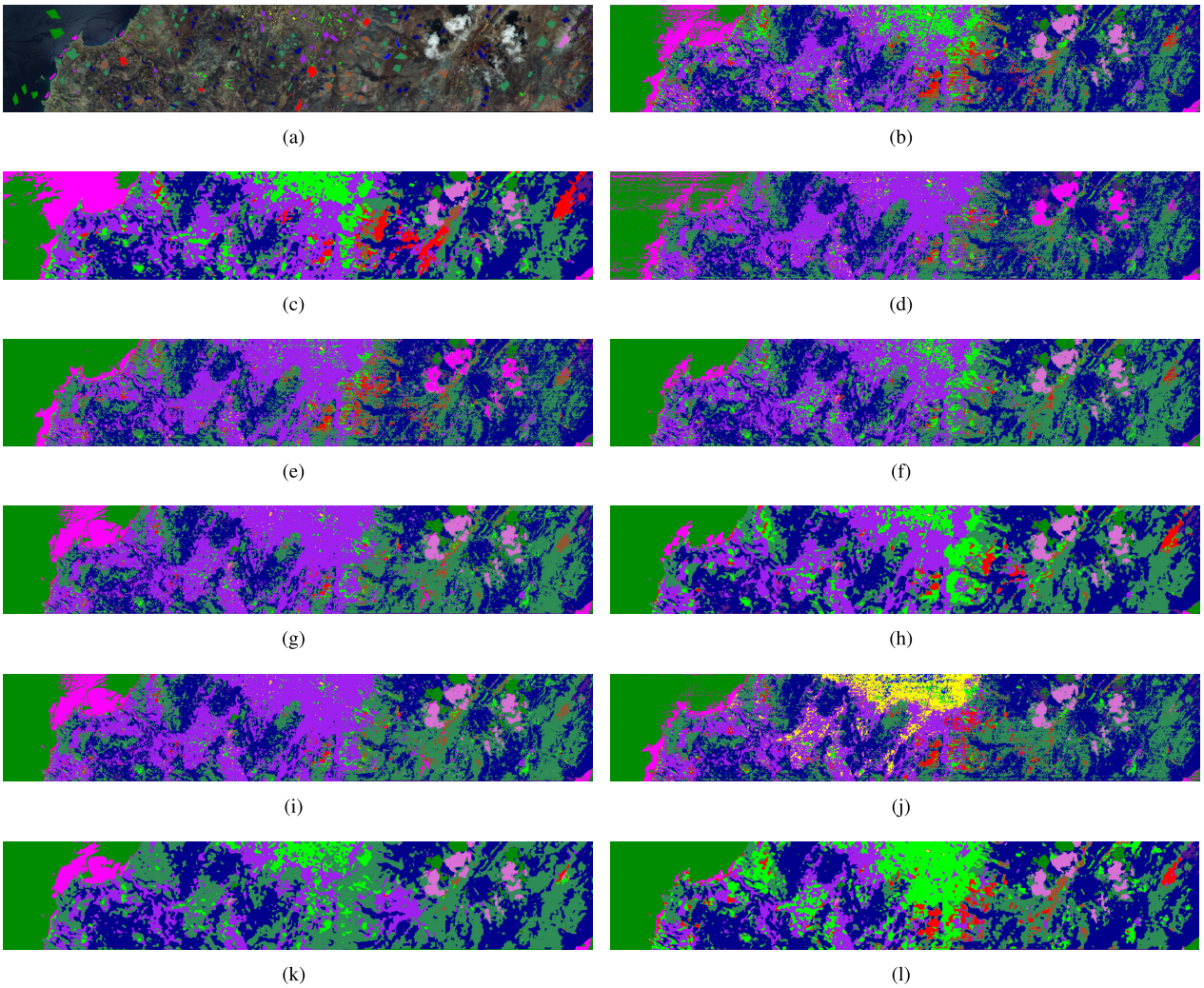


Fig. 10. Full classification maps obtained by different methods on Loukia→Dioni. (a) False-color image of Dioni. (b) DNN. (c) GCN. (d) DANN. (e) D-CORAL. (f) DAN. (g) MADA. (h) JCGNN. (i) CDA. (j) BCDM. (k) AGAN. (l) SILDA-VC.

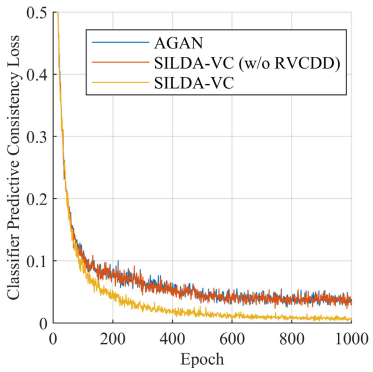


Fig. 11. CPC loss on BOT5→BOT6.

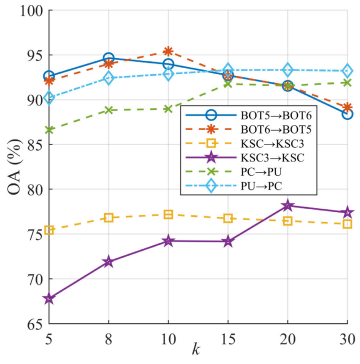


Fig. 12. OAs achieved by SILDA-VC under different k's.

Small loss suggests high predictive consistency between the real and virtual classifiers. Taking BOT5→BOT6 as an example, the CPC losses of AGAN, SILDA-VC (w/o RVCDD), and SILDA-VC are printed out, as shown in Fig. 11. It can be seen that CPC losses obtained by AGAN and SILDA-VC (w/o RVCDD) are significantly higher than those by SILDA-VC, which indicates that SILDA-VC can make the class of the target-domain sample assigned by the classifier as consistent as possible with class of the most similar source-domain class prototype.

D. Parameter Analysis

In this section, we analyze the effects of several important parameters (k, ϵ_{RV}, and ϵ_{SPC}) in SILDA-VC; taking

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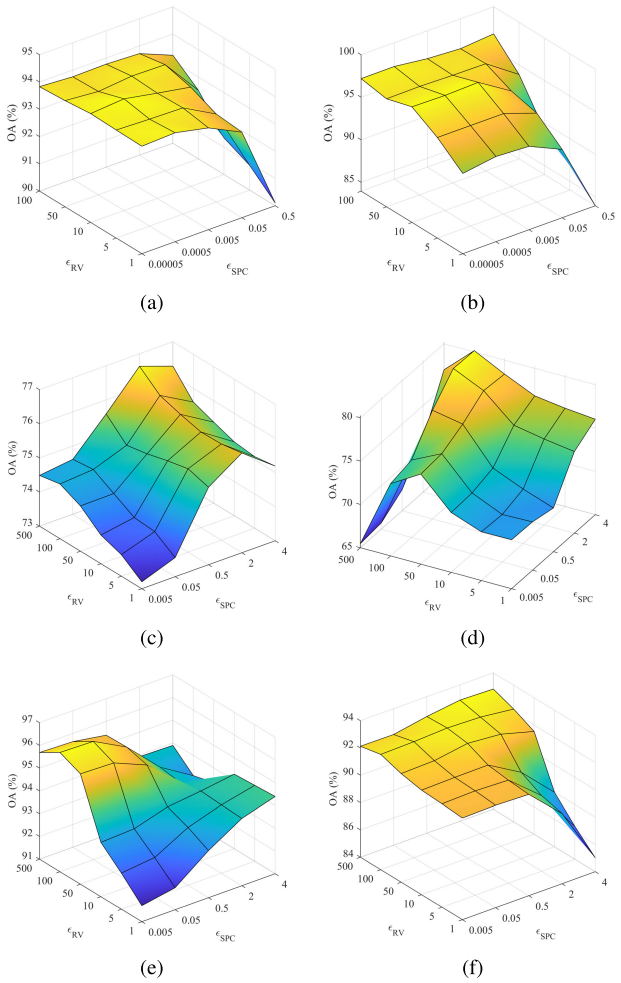


Fig. 13. OAs achieved by SILDA-VC under different ϵ_{RV}'s and ϵ_{SPC}'s. (a) BOT5→BOT6. (b) BOT6→BOT5. (c) KSC→KSC3. (d) KSC3→KSC. (e) PC→PU. (f) PU→PC.

BOT5→BOT6, BOT6→BOT5, KSC→KSC3, KSC3→KSC, PC→PU, and PU→PC as examples, the OAs obtained by SILDA-VC under different values are shown in Figs. 12 and 13. It can be seen from Figs. 12 and 13 that the following conditions hold.

1) As k increases, OA increases first and then decreases. Too small k will lead to the inadequate representation of manifold relationships among samples; too large k will introduce the noisy neighbors around the selected samples. OA reaches or approaches the highest value when $k = 10$.
2) OA increases and then decreases with increasing ϵ_{RV} and ϵ_{SPC}. Too large ϵ_{RV} or ϵ_{SPC} will make $\mathcal{L}_{RVCDD}$ and $\mathcal{L}_{SPC}$ take a large proportion in the entire experimental process. Too small ϵ_{RV} will make the hidden-layer features contain much redundant information, and the instance-level adaptation on this basis will not be effective. Too small ϵ_{SPC} will lead to inadequate adaptation. OA reaches or approaches the highest value when $\epsilon_{RV} = 100$ and $\epsilon_{SPC} = 2$.

IV. Conclusion

SILDA-VC is proposed for unsupervised HSI classification in this article, which includes a real classifier and a virtual classifier. By minimizing the RVCDD loss, the source- and target-domain samples with similar features are encouraged to be classified into the same class, thus enhancing the discriminability of features. Then, a soft instance-level domain adaptation method is designed to adapt the source- and target-domain samples softly in an instance-level way. SILDA-VC can not only keep similar samples close and push dissimilar samples away in the feature space but also increase the distinguishability of output features of classifier, keeping the samples away from their respective decision boundaries. Compared with several recently proposed domain adaptation methods, SILDA-VC achieves the best performance on different HSI transfer tasks. It should be noted that SILDA-VC is only applicable for the cross-domain classification task under the condition that the source and target domains have the same classes of ground objects. However, due to differences in shooting sensors and regions, the source and target domains tend to be heterogeneous and contain different classes of ground objects. In the future, we consider adopting meta-learning to relax the class consistency requirement of SILDA-VC.

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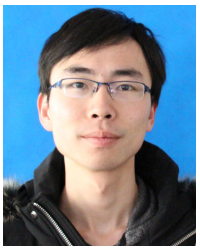
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